FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – II

JEE (Advanced)-2019

PAPER – 1

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-D.**

1. ***Section-A (01– 06, 19 – 24, 37 - 42)*** *contains 18 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen, both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*

   *Negative Marks : **–2** In all other cases.*

   Section-A (07 – 10, 25 – 28, 43 - 46) *contains 12 questions. Based on this section contains* ***TWO (02)*** *paragraphs. Based on each paragraph, there are* ***TWO (02)*** *questions.*
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-D (11 – 18, 29 – 36, 47 – 54)*** *contains 24 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

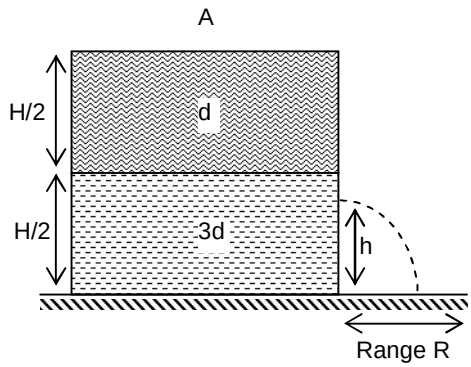
1. Two particles A and B, each carrying a charge Q, are held fixed with a separation d between them. A particle C having a mass m and charge –q, in kept at the middle point of line AB.
It displaced through a distance x perpendicular to AB. Assume x << d. Then:
(A) Force experienced by C is proportional to x
(B) Force experienced by C is proportional to d
(C) Particle C may execute SHM with time period $\left(\frac{m\pi^3\varepsilon_0 d^3}{qQ}\right)^{1/2}$
(D) Particle C may executes SHM with time period $\left(\frac{m\pi^2\varepsilon_0 d^2}{qQ}\right)^{1/2}$

2. If the potential difference of Coolidge tube producing X-ray is increased, then choose the correct option(s).
(A) the interval between $\lambda_{k\alpha}$ and $\lambda_{k\beta}$ increases
(B) the interval between $\lambda_{k\alpha}$ and λ_0 increases
(C) the interval between $\lambda_{k\beta}$ and λ_0 increases
(D) λ_0 does not change
Here λ_0 is cut off wavelength and $\lambda_{k\alpha}$ and λ_{kB} are wavelength of k_α and k_B characteristic X-rays.

3. For a single material
(A) its absorptive power and emissivity, have equal magnitude
(B) molar specific heat capacity is proportional to its specific heat capacity.
(C) mass is proportional to number of moles.
(D) density is proportional to volume of material.

Space for Rough work

4. A container of large uniform cross sectional area A resting on a horizontal surface holds two immiscible non viscous and incompressible liquids of density d and 3d, each of height H/2. The lower density liquid is open to the atmosphere having pressure P_0. A tiny hole of area a(a<<A) is punched to the vertical side of lower container at a height h (0<h<H/2) for which range is maximum.



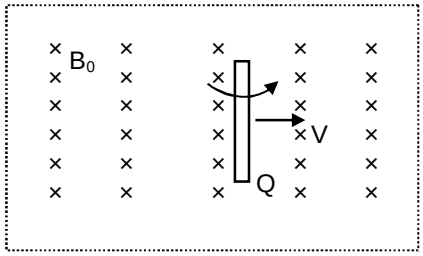
(A) h = H /3

(B) Range R = $\frac{2H}{3}$

(C) Range R = $\frac{3H}{2}$

(D) Velocity of efflux $v = \sqrt{\frac{2}{3}gH}$

5. A uniform thin rod PQ of length 'L' is moving in a uniform magnetic field B_0 such that velocity of its centre of mass is v and angular velocity is $\omega = \frac{4v}{L}$. Then



(A) e.m.f. between end P and end Q of the rod is $B_0 l v$

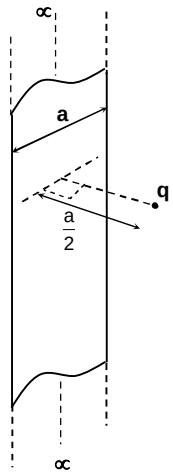
(B) end P of the rod is at higher potential than end Q of the rod

(C) end Q of the rod is at higher potential than end P of the rod

(D) the electrostatic field induced in the rod has same direction at all points along the length of rod.

Space for Rough work

6. There is an infinite long straight surface having width 'a'. A point charge q is placed at a perpendicular distance $\frac{a}{2}$ from the surface symmetrically as shown in the figure. If the flux linked with this infinitely long surface due to charge q is $\frac{q}{k\varepsilon_0}$. Then find the value of k is



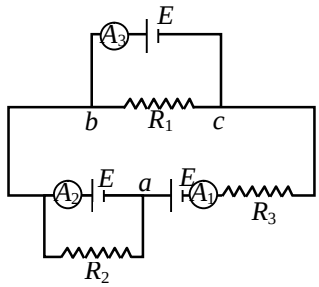
(A) 6 (B) 4
(C) 2 (D) 12

(Paragraph Type)

This section contains **TWO** paragraphs. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 7 and 8

In the circuit shown in figure, $R_1 = R_2 = R_3 = 3\Omega$ and e.m.f. of each cell is E = 4V and negligible internal resistance. All ammeters are ideal. Match the following:

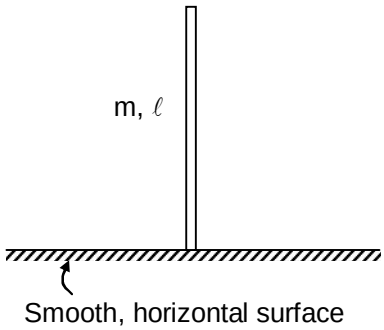


7. Reading of ammeter A_1 in ampere is
(A) 2/3 (B) 4/3
(C) 1 (D) 4

Space for Rough work

8. Reading of ammeter A_2 in ampere is
(A) 8/3 (B) 1/3
(C) 1 (D) 2

Paragraph for Question Nos. 9 and 10



9. The thin rod is released from the vertical position as shown, and falls by itself, then angular speed of the rod just before it falls flat, will be
(A) $\sqrt{\frac{g}{\ell}}$ (B) $\sqrt{\frac{3g}{\ell}}$
(C) $\sqrt{\frac{4g}{\ell}}$ (D) $\sqrt{\frac{6g}{\ell}}$

10. KE of the rod, just before it falls flat, will be
(A) $mg\ell$ (B) $mg\frac{\ell}{2}$
(C) $mg\frac{\ell}{4}$ (D) $mg\frac{\ell}{6}$

Space for Rough work

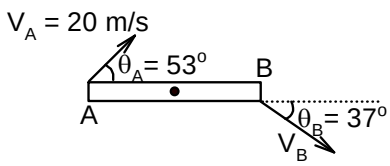
SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

11. A 10 kg mass rests on a horizontal frictionless table. How much energy (in Joule) is needed to accelerate the mass from rest to a speed of 5 m/s?

12. A block of mass 2kg is gently placed over a massive plank moving horizontally over a smooth surface with velocity 6 m/s. The coefficient of friction between the block and plank is 0.2. The distance travelled by the block till it slides on the plank is ($g=10\ m/s^2$)

13. The two ends A and B of a uniform rod of length ℓ = 1m are moving with velocities V_A and V_B as shown. The velocity V_B (m/s) is

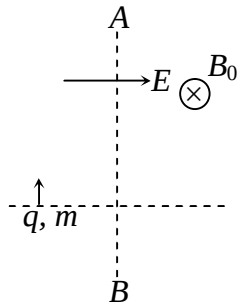


14. A hollow sphere and a solid sphere have equal mass and equal moment of inertia about the respective diameter. Find the ratio of their square of radii is given by

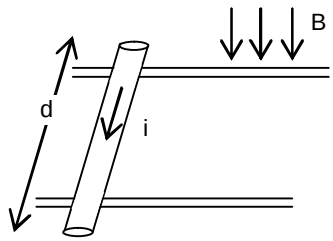
15. A steel wire of cross-sectional area $2mm^2$ and Young's modulus $2\times10^{11}N/m^2$ is stretched longitudinally by a force of 200 N. The elastic potential energy stored per unit volume in the string is $n \times 10^4$. Find the value of 'n'.

Space for Rough work

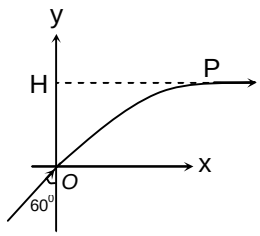
16. A uniform electric field *E* is present horizontally along the paper throughout region but uniform magnetic field B_0 is present horizontally (perpendicular to plane of paper in inward direction) right to the line *AB* as shown. A charge particle having charge *q* and mass *m* is projected vertically upward and crosses the line *AB* after time t_0. The speed of projection if particle moves after t_0 with constant velocity. (given *qE* = *mg*) is nt_o. Find the value of 'n'. ($g = 9.8\ m/s^2$)



17. A cylindrical uniform rod of mass 0.72 kg and radius 6 cm rests on two parallel rails, that are d = 50 cm apart. The rod caries a current I = 48A (In the direction shown) and rolls along the rails without slipping. If it starts from rest, uniform magnetic field of magnitudes 0.25 T is directed perpendicular to the rod and the rail, then find the friction force(In N) between rod and rails



18. A system of coordinates is drawn in a medium whose refractive index varies as $\mu = \frac{2}{1+y^2}$, where $0 \le y \le 1$. A ray of light is incident at origin at an angle 60^0 with *y* –axis as shown in the figure. At point P ray becomes parallel to x-axis. Find the value of 'H'.



Space for Rough work

Chemistry

PART – II

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. Choose correct statement(s) regarding PCl_5 molecule.
(A) Phosphorus uses 3d orbital in hybridization
(B) Three different bond angles are observed in the molecule
(C) In solid state it contains ion-pairs
(D) All P – Cl bonds are of equal length

20. Which of the following solution(s) form buffer if they are taken in 2 : 1 molar ratio?
(A) CH_3COOH and CH_3COONa (B) NH_4OH and NaCl
(C) NH_4OH and NH_4Cl (D) $NaHCO_3$ and Na_2CO_3

21. Which of the following compound(s) on catalytic hydrolysis forms carbonyl compound(s)?
(A) $CH_3CH = CH_2$ (B) $CH_3C \equiv CH$
(C) $CH_3C \equiv CCH_3$ (D) $HC \equiv CH$

22. Which of the following compound(s) contain(s) hydrogen atoms?
(A) Silicate anions (B) Silicones
(C) Silanes (D) Selenes

23. Which of the following complex(es) is/are insoluble in water?
(A) $Na_2[Zn(CN)_4]$ (B) $[Co(NH_3)_3Cl_3]$
(C) $[Cr(H_2O)_6]Br_3$ (D) $H_2[SiF_6]$

24. $FeCl_3 + K_4\left[Fe(CN)_6\right] \longrightarrow$ Precipitate $\downarrow$
The correct statement(s) regarding the precipitate in above reaction is/are
(A) it is blue in colour
(B) one mole of the precipitate can consume 12 moles of NaOH
(C) the maximum van't Hoff factor of the ppt. can be seven
(D) the ppt. becomes soluble in hot water

Space for Rough work

(Paragraph Type)

This section contains **TWO** paragraphs. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 25 and 26

For dissociation reactions, like $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$, the degree of dissociation of SO_3 can be determined by vapour density measurement of the reacting species through the following relation.

$$\alpha = \frac{D - d}{(n-1)d}$$

The molecular mass of the reaction mixture at equilibrium is 60. Initially only SO_3 was taken for the reaction.

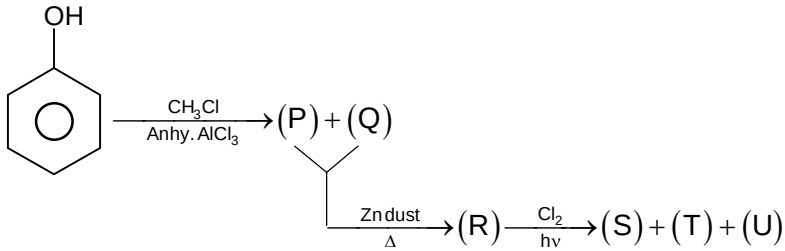
Answer the following questions on the basis of above write up.

25. What is the value of 'D' in the reaction?
(A) 60 (B) 30
(C) 40 (D) 80

26. What is the value of 'α' for the reaction?
(A) 0.2 (B) 0.4
(C) 0.6 (D) 0.8

Space for Rough work

Paragraph for Question Nos. 27 and 28



(S) $\rightarrow$ Monosubstituted product
(T) $\rightarrow$ Disubstituted product
(U) $\rightarrow$ Trisubstituted product
Answer the following questions on the basis of above write up.

27. The unknown compounds S, T, U are
(A) position isomers (B) functional isomers
(C) metamers (D) different compounds

28. The most deactivating unknown compound towards EAS reaction is:
(A) R (B) U
(C) T (D) S

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

29. The energy of the first orbit of hydrogen atom is -13.6 eV. If the fourth ionization enthalpy of beryllium atom is expressed as $Y \times 10^{-19}$ J, what is the value of 'Y'?

30. How much heat is absorbed in calorie unit by heating 5 moles of a gas at constant pressure from 900 K to 1727.21 K?
[C_P = 4R, R = 2 Cal]

Space for Rough work

31. What is the kinetic energy of one mole of helium gas at 10000 K in kJ mol^{-1} unit?

32. Optically active chloride $(P) \xrightarrow{KOH} (Q)$

$(Q) \xrightarrow[Heat]{Conc.H_2SO_4} \underset{Major}{(R)} + \underset{Minor}{(S)}$

$(R) \xrightarrow{Ozonolysis} (T) + (U)$

'Q' forms a hydrocarbon when treated with C_2H_5MgBr. 'U' is the simple aliphatic aldehdye that give aldol condensation. (T) is acetophenone. What is the molar mass of (P) in gram unit?

33. $(X) \xrightarrow[Decarboxylation]{Heating} CH_3 - \overset{\overset{O}{\|}}{C} - CH_3 + 3CO_2$

One mole of (X) on heating produces one mole of acetone and three moles of CO_2. The tri-sodium salt of the keto acid (X) reacts with CH_3COCl in presence of pyridine to form acid anhydride. How much CH_3COCl in gram can be completely consumed by one mole of the tri-sodium salt of (X)?

34. $\underset{\text{metal chloride}}{(P)} \xrightarrow{NaOH} \underset{\text{green ppt.}}{(Q)\downarrow} \xrightarrow{\text{Excess NaOH}} \underset{\text{green solution}}{(R)}$

$\underset{\text{Soln}}{(R)} \xrightarrow{H_2O_2} \underset{\text{yellow soln}}{(S)} \xrightarrow[BaCl_2]{CH_3COOH} \underset{\text{yellow ppt.}}{(T)\downarrow}$

What is the molar mass of (P) in gram unit?

35. Electrolysis of molten $CuSO_4$ is carried out in an electrolytic cell, using inert electrodes. How many gram of copper will be deposited at cathode by passing 14 F of electricity through the cell?

36. How many gram of magnesium carbonate ($MgCO_3$) can be completely dissolved in 3.26 litre of 5 M HCl solution according to the following reaction?

$MgCO_3 + 2HCl \longrightarrow MgCl_2 + CO_2 + H_2O$

Space for Rough work

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. If $f(x) = 2x - \sin x$ and $g(x) = \sqrt[3]{x}$, then

(A) range of gof is R
(B) gof is one – one
(C) both f and g are one – one
(D) both f and g are onto

38. $\lim\limits_{x\to\infty} \sqrt{x}\left(\sqrt{x+1} - \sqrt{x}\right)$ equals

(A) $\lim\limits_{x\to 0} \dfrac{\ln(1+x) - x}{x^2}$
(B) $\lim\limits_{x\to 0} \dfrac{1-\cos x}{x^2}$
(C) $\lim\limits_{x\to 0} \dfrac{\sqrt{1+x} - 1}{x}$
(D) $\lim\limits_{x\to 0} \dfrac{\sqrt{x}}{\sqrt{x+\sqrt{x^2+2x}}}$

39. If $f(x) = \begin{cases} \int_0^x \left(1 + |1-t|\right) dt, & x > 2 \\ 5x + 1, & x \le 2 \end{cases}$, then

(A) f(x) is not continuous at x = 2
(B) f (x) is continuous but not differentiable at x = 2
(C) f (x) is differentiable everywhere
(D) the right derivative of $f(x)$ at x = 2 does not exist

40. If $f(x) = \begin{vmatrix} n & n+1 & n+2 \\ {}^{n}P_{n} & {}^{n+1}P_{n+1} & {}^{n+2}P_{n+2} \\ {}^{n}C_{n} & {}^{n+1}C_{n+1} & {}^{n+2}C_{n+2} \end{vmatrix}$, where the symbols have their usual meanings, then $f(x)$ is divisible by

(A) $n^2 + n + 1$
(B) $(n+1)!$
(C) $n!$
(D) $n^2 + 3n + 1$

Space for Rough work

41. The solution/s of the equation $9\cos^{12} x + \cos^2 2x + 1 = 6\cos^6 x \cos 2x + 6\cos^6 x - 2\cos 2x$ is/are

(A) $x = n\pi + \frac{\pi}{2}, n \in I$ (B) $x = n\pi \pm \cos^{-1}\sqrt[4]{\frac{2}{3}}, n \in I$

(C) $x = n\pi \pm \cos^{-1}\sqrt{\frac{2}{3}}, n \in I$ (D) $x = n\pi, n \in I$

42. If a straight line through a point $P(\alpha, 2), \alpha \neq 0$, meets the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ at A and D and x – and y–axes at B and C respectively such that $PA \times PD = PB \times PC$, then '$\alpha$' can lie in the interval

(A) $[7, \infty)$ (B) $(-12, -8)$

(C) $(-5, 0)$ (D) $(10, \infty)$

(Paragraph Type)

This section contains **TWO** paragraphs. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 43 and 44

Let $a_1, a_2, a_3, \ldots$ be a harmonic progression such that $a_1 = 5$ and $a_{20} = 25$.

43. The value of n for which a_n is maximum is

(A) 22 (B) 23

(C) 24 (D) 25

44. The least value of n for which $a_n < 0$ is

(A) 22 (B) 23

(C) 24 (D) 25

Space for Rough work

Paragraph for Question Nos. 45 and 46

Let $A(4,-2)$ and $B(0,1)$ be two points on a parabola, the normals at which intersect at the point $P(2,2)$.

45. The equation of the directrix of the parabola is

(A) $x+y+1=0$ (B) $x=2$

(C) $y=-3$ (D) $2x+y-1=0$

46. The focus of the parabola is

(A) $\left(\frac{6}{5},-\frac{6}{5}\right)$ (B) $\left(\frac{16}{5},-\frac{7}{5}\right)$

(C) $\left(\frac{12}{5},\frac{6}{5}\right)$ (D) $\left(-\frac{6}{5},\frac{6}{5}\right)$

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

47. If $2<x^2<3$, then the positive root of $\left\{x^2\right\}=\left\{\frac{1}{x}\right\}$, (where {x} denotes the fractional part of x) is equal to

48. Let $L=\lim_{x\to 0}\left(\left[\frac{100x}{\sin x}\right]+\left[\frac{99\sin x}{x}\right]\right)$, where [.] denotes the greatest integer function. The number of divisors of L is equal to

Space for Rough work

49. If $f(x)+f(y)=f\left(\dfrac{x+y}{1-xy}\right)$ for all $x,y\in R$, $xy\neq 1$ and $\lim\limits_{x\to 0}\dfrac{f(x)}{x}=2$, then the value of $\dfrac{f\left(\sqrt{3}\right)}{f'(-2)}$ is

50. The number of solutions of the equation $\sin\left(\dfrac{x}{2}\right)+2\pi x=x^2+\pi^2+1$ is

51. The minimum value of $(\tan A-\cos B)^2+(\cot A-\sin B)^2$, where A and B are independent variables, is of the form $a-b\sqrt{c}$, where a, b, c are natural numbers. The value of $a+b+c$ is equal to

52. The smallest value of 'a' for which the inequality $(a-1)x^2-(a+1)x+(a-1)\geq 0$ holds for all $x\geq 2$, is equal to

53. Three distinct dice are thrown and the sum of the numbers appearing on the top faces is 12. If n is the number of ways in which this is possible, the sum of the digits of n is equal to

54. The papers of 4 students can be checked by any one of the 7 teachers. If the probability that all the 4 papers are checked by exactly 2 teachers is A, then the value of 50A is equal to

Space for Rough work